

Ahmad Nawaz

Senior AI/ML Engineer | TensorFlow | PyTorch | NLP | Computer Vision

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PROFESSIONAL SUMMARY

Senior AI/ML Engineer with 9 years of experience building intelligent systems that drive business efficiency and deliver actionable insights. I specialize in predictive modeling, natural language processing, and computer vision using TensorFlow, PyTorch, and Scikit-Learn. I have designed and deployed end-to-end AI workflows, production-ready AI systems with Docker and Kubernetes, and data-driven solutions across finance, healthcare, and retail. My work focuses on solving real-world problems, improving workflows, and creating scalable AI solutions that enhance user experience and business performance. I am passionate about applying AI to transform operations and deliver measurable impact for organizations.

SKILLS

Machine Learning & AI Frameworks	Python, R, TensorFlow, PyTorch, Scikit-Learn, Keras, Hugging Face, LangChain, LangGraph, LlamaIndex, Agentic AI, Crew AI, Claude AI, RAG, AutoML, OpenCV, NLTK, BERT, GANs, CNN, RNN, ResNet, UNet, GCN, GraphSAGE, XGBoost, LightGBM, CatBoost, Reinforcement Learning, LLM Orchestration, Prompt Engineering, GenAI, Deep Learning Models (CNN, RNN, ResNet, UNet), Multi-agent Systems, PyTorch Lightning, DeepSpeed, Ray, FAISS, Weights & Biases.
Data Visualization & Frontend Tools	Plotly, Dash, React (Dashboards), Next.js, TypeScript, n8n, JavaScript, D3.js, Bokeh, Matplotlib, Seaborn, Three.js, Vector Databases (Pinecone, Chroma)
Data Engineering & Cloud Platforms	MongoDB, MySQL, PostgreSQL, Cassandra, Neo4j, Redis, Supabase, Firebase, Databricks, Snowflake, Docker, Kubernetes, Terraform, AWS (S3, Lambda, EC2, SageMaker, Bedrock), Google Cloud ML Engine, ETL (Python, SQL, Airflow), CI/CD (Jenkins, GitLab, GitHub Actions) GCP, Azure ML, MLflow, Triton Inference Server, Airflow, Prefect, AI Observability Tools (OpenTelemetry, LangSmith), Data Privacy & Security

WORK EXPERIENCE

SoftServe | Austin, TX (Remote)

Senior Software Engineer - AI/ML

January 2022 - October 2025

SoftServe delivers AI-driven consulting and software solutions to healthcare, finance, retail, and media clients. I led projects designing intelligent systems and scalable AI models to drive business efficiency and insight generation. Engineered multi-agent AI systems using CrewAI to simulate strategic decision-making for operational automation.

Responsibilities:

- Led a team of 6-15 engineers to develop Python-based AI systems for healthcare, finance, retail and others.
- Developed multi-agent AI systems with CrewAI in Python for decision-making and workflow automation.
- Deployed personalized recommendation models using TensorFlow, PyTorch, and AWS SageMaker in Python.
- Developed ETL pipelines in Python and SQL for streamlined data processing and cloud integration.
- Implemented CI/CD pipelines with Jenkins and GitLab for automated testing and deployment.
- Implemented data preprocessing pipelines in Python with Pandas and NumPy, improving training data quality.
- Enhanced NLP models using BERT and SpaCy in Python for improved chatbot performance and sentiment analysis.
- Designed scalable machine learning pipelines on Google Cloud ML Engine, leveraging Python for orchestration.
- Automated the ML lifecycle with AWS SageMaker, streamlining model deployment in Python.
- Built image preprocessing workflows with OpenCV in Python for computer vision tasks.
- Applied Python with NLTK for sentiment analysis and keyword extraction from user feedback.
- Integrated RAG (Retrieval-Augmented Generation) in Python for efficient document retrieval and AI processing.
- Developed real-time anomaly detection models in Python using Azure Cognitive Services for fraud mitigation.
- Optimized models with hyperparameter tuning using AWS SageMaker and Google Cloud AI in Python.
- Built and deployed real-time AI services in Python using FastAPI for fast model inference.

- Collaborated on LLM orchestration solutions with LangChain using Python.
- Utilized Graph Neural Networks (GCN, GraphSAGE) for social network analysis and relationship predictions.
- Integrated Python with n8n for workflow automation, streamlining system integration.
- Developed model evaluation frameworks using Matplotlib and Seaborn for visualizing performance metrics.
- Handled SQL (PostgreSQL) and NoSQL (MongoDB) databases for model training and data management.
- Delivered AI-driven insights and models using Python, driving business efficiency for client platforms.
- Mentored junior engineers in Python and machine learning best practices.

Tech Used: Python, R, TensorFlow, PyTorch, Scikit-Learn, Pandas, NumPy, Keras, Google Cloud ML Engine, ETL Pipelines, CI/CD Pipelines, AWS Bedrock, AWS SageMaker, OpenCV, NLTK, SQL, NoSQL, Jupyter Notebooks, Docker, Apache Spark, Azure Cognitive Services, n8n, Matplotlib, Seaborn, RAG, REST APIs, Java, Golang, SciPy, FastAPI, LangChain, LlamaIndex, Graph Neural Networks, PyTorch Lightning, DeepSpeed, Ray.

Mercury Development | Miami, FL (Remote)

Senior Software Engineer - AI/ML

March 2019 - December 2021

Mercury Development provides AI/ML, predictive analytics, and automation solutions for web, mobile, and enterprise clients. I delivered data-driven models, recommendation systems, and intelligent applications across multiple industries. Engineered anomaly detection models using AutoML and XGBoost to optimize supply chain operations efficiently.

Responsibilities:

- Developed and deployed BERT-based and Dialogflow-powered chatbots to enhance conversational understanding, improve customer engagement, and streamline service operations.
- Combined BERT, Speech-to-Text APIs, and Agentic AI to enable hands-free mobile application controls.
- Scaled ML applications using AWS Lambda serverless functions for faster, reliable inference processing.
- Developed MATLAB-based predictive maintenance models, reducing manufacturing equipment downtime.
- Implemented XGBoost credit scoring systems to accurately assess borrower risk and optimize lending decisions.
- Applied Azure ML to healthcare datasets, delivering actionable predictive insights for treatment planning.
- Built collaborative filtering recommendation engines on Apache Mahout, boosting e-commerce sales by 20%.
- Integrated facial emotion recognition using Microsoft Cognitive Toolkit to analyze engagement patterns.
- Designed and deployed Quantum ML-based traffic prediction models to enhance navigation efficiency.
- Forecasted financial market trends using Statsmodels and time-series analysis to support strategic planning.
- Developed decision tree-based real-time ad bidding systems to maximize marketing ROI for campaigns.
- Utilized graph neural networks to analyze social network data, identifying key influencers and communities.

Tech Used: AutoML, XGBoost, BERT, Reinforcement Learning, MATLAB, Azure ML, Apache Mahout, Microsoft Cognitive Toolkit, Speech-to-Text APIs, Agentic AI, Statsmodels, Decision Trees, Deep Learning, Graph Neural Networks, Dialogflow, REST APIs, Java, JavaScript.

West Agile Labs | San Francisco, CA

Software Engineer

August 2016 - February 2019

West Agile Labs is a digital product development agency that specializes in delivering AI-driven web applications and scalable solutions for businesses. They focus on creating impactful, user-centric platforms that drive business efficiency and innovation.

Responsibilities:

- Developed AI web applications using Python, TensorFlow, and PyTorch for client-specific solutions.
- Built responsive web interfaces with React.js for interactive ML visualizations.
- Created REST APIs with Flask and Node.js to integrate ML models into web apps.
- Processed and cleaned data with Pandas and NumPy for accurate ML results.
- Implemented computer vision with OpenCV for automated image analysis.
- Deployed AI solutions in Docker containers for consistent and portable execution.
- Built recommendation systems and dashboards to enhance business intelligence.
- Integrated databases (SQL, MongoDB) to manage and store ML data.

Tech Used: Python, JavaScript, React, Node.js, TensorFlow, PyTorch, Scikit-Learn, Pandas, NumPy, OpenCV, Flask, REST APIs, SQL, MongoDB, Docker.

EDUCATION

Bachelor Degree in Software Engineering (2016)

Air University, PK